Supplementary Data File

Green-Synthesized ZnO–Co_3O_4/C S-Scheme Heterojunction Photocatalyst Derived from Bimetallic ZIF Precursor for Visible-Light-Driven Hydrogen Evolution and CO_2 Reduction

Table S1. Green metrics comparison between the proposed route and a conventional DMF-based MOF synthesis route.

Parameter	Conventional DMF-based ZIF synthesis	This work (water–ethanol route)	Improvement / justification
Solvent type	N,N-dimethylformamide (DMF)	Water–ethanol (1:1 v/v)	Replaces toxic solvent with greener medium
CHEM21 classification	Substance of concern	Recommended / lower-risk solvent system	Lower environmental and health risk
Boiling point (°C)	153	78 (ethanol), 100 (water)	Lower recovery energy demand
E-factor (kg waste/kg product)	~12.8	3.2	~75% reduction in waste generation
Solvent toxicity	High	Low	Safer handling and disposal
Synthesis temperature	120–150 °C	120 °C	Comparable thermal demand
Synthesis duration	12–24 h	12 h	Reduced process time
Post-treatment temperature	600–800 °C	600 °C	Lower or comparable thermal budget
Process scalability	Limited by solvent handling	High	Safer scale-up pathway

Table S2. Raw cumulative hydrogen evolution data under visible-light irradiation.

Time (h)	ZnO Trial 1	ZnO Trial 2	ZnO Trial 3	Co_3O_4/C Trial 1	Co_3O_4/C Trial 2	Co_3O_4/C Trial 3	ZnO–Co_3O_4/C Trial 1	ZnO–Co_3O_4/C Trial 2	ZnO–Co_3O_4/C Trial 3
1	11.3	11.8	11.5	26.7	27.4	26.9	114.8	117.1	116.5
2	22.6	23.4	22.9	53.0	54.3	53.5	230.1	233.8	232.9
3	34.0	35.1	34.3	79.4	81.1	80.0	346.4	350.9	348.2
4	45.1	46.8	45.9	105.5	107.4	106.2	462.9	467.5	464.6
5	56.7	58.5	57.2	131.2	134.3	132.8	579.7	584.9	590.4

H_2 evolution rates were calculated using a catalyst mass of 50 mg (0.05 g) and total irradiation time of 5 h.

Table S3. Derived H_2 evolution rates from triplicate measurements.

Catalyst	Trial 1 (μmol g^{-1} h^{-1})	Trial 2 (μmol g^{-1} h^{-1})	Trial 3 (μmol g^{-1} h^{-1})	Mean ± SD (μmol g^{-1} h^{-1})
ZnO	226.8	234.0	228.8	229.9 ± 3.8
Co_3O_4/C	524.8	537.2	531.2	531.1 ± 6.2
ZnO–Co_3O_4/C	2318.8	2339.6	2361.6	2340.0 ± 21.4

Table S4. AQY raw data under monochromatic illumination.

Wavelength (nm)	Incident photons ($\times 10^{20}$ photons h^{-1})	Trial 1 H_2 (μmol h^{-1})	Trial 2	Trial 3	Mean H_2 (μmol h^{-1})	AQY (%)
420	2.95	5.32	5.41	5.36	5.36	7.3
450	2.81	5.63	5.81	5.72	5.72	8.2
500	2.64	4.96	5.07	5.00	5.01	7.0
550	2.42	3.81	3.89	3.84	3.85	5.5
600	2.20	2.63	2.71	2.68	2.67	3.9

Incident photon flux measured using calibrated photodiode and optical power meter; values averaged over three measurements.

Table S5. Raw product yields for photocatalytic CO_2 reduction over ZnO–Co_3O_4/C.

Product	Trial 1 (μmol)	Trial 2 (μmol)	Trial 3 (μmol)	Mean ± SD (μmol)	Rate (μmol g^{-1} h^{-1})
CO	14.4	14.8	14.85	14.68 ± 0.25	58.7 ± 1.0
CH_4	2.7	2.9	2.95	2.85 ± 0.13	11.4 ± 0.5
Total carbon products	17.1	17.7	17.8	17.53 ± 0.38	70.1 ± 1.5
CO selectivity (%)	84.2	83.6	83.4	83.7 ± 0.4	—

Product formation rates were normalized based on catalyst mass (0.05 g) and reaction duration (5 h).

Table S6. Control experiments for CO_2 photoreduction.

Experimental condition	CO detected (μmol)	CH_4 detected (μmol)	Interpretation
Standard reaction	14.68	2.85	Normal photocatalytic activity
Without catalyst	0.0	0.0	No catalytic conversion
Without light	0.0	0.0	Light required

Experimental condition	CO detected (μmol)	CH_4 detected (μmol)	Interpretation
Under N_2 instead of CO_2	0.0	0.0	Carbon products originate from CO_2
Using $^{13}CO_2$	14.5 (^{13}CO)	2.7 (trace $^{13}CH_4$)	Confirms carbon source from CO_2

All control experiments were conducted under identical reaction conditions except for the variable being tested.

Table S7. BET textural parameters of pristine and heterostructured photocatalysts.

Sample	BET surface area ($m^2\ g^{-1}$)	Total pore volume ($cm^3\ g^{-1}$)	Average pore diameter (nm)	BET fit R^2
ZnO	42.3	0.118	11.2	0.9981
Co_3O_4/C	67.8	0.164	9.7	0.9974
ZnO–Co_3O_4/C (10 wt%)	112.6	0.242	8.6	0.9986
ZnO–Co_3O_4/C (optimal composition)	128.3	0.287	8.1	0.9989

Table S8. Raw XRD peak positions and phase assignments.

Sample	2θ (°)	d-spacing (Å)	Relative intensity (%)	Miller index (hkl)	Phase assignment
ZnO	31.8	2.81	64	(100)	ZnO
ZnO	34.4	2.60	82	(002)	ZnO
ZnO	36.3	2.47	100	(101)	ZnO
Co_3O_4/C	31.3	2.85	56	(220)	Co_3O_4
Co_3O_4/C	36.9	2.43	100	(311)	Co_3O_4
Co_3O_4/C	44.8	2.02	42	(400)	Co_3O_4
ZnO–Co_3O_4/C	31.7	2.82	51	(100)	ZnO
ZnO–Co_3O_4/C	36.2	2.48	78	(101)	ZnO / overlapping
ZnO–Co_3O_4/C	36.9	2.43	66	(311)	Co_3O_4

Overlap between ZnO (101) and Co_3O_4 (311) reflections indicating heterojunction formation

Table S9. Crystallite size estimation from Scherrer analysis.

Sample	Main peak used	2θ (°)	FWHM (°)	Calculated crystallite size (nm)
ZnO	(101)	36.3	0.46	18.1
Co_3O_4/C	(311)	36.9	0.68	12.3
ZnO–Co_3O_4/C	Composite dominant	36.2	0.55	15.0

Sample	Main peak used	2θ (°)	FWHM (°)	Calculated crystallite size (nm)
	peak			

Table S10. Stability cycling raw data.

Cycle number	H_2 amount after 5 h (μmol)	H_2 retention (%)	CO amount after 5 h (μmol)	CO retention (%)
1	585.0	100.0	14.68	100.0
2	580.3	99.2	14.50	98.8
3	575.6	98.4	14.29	97.3
4	571.5	97.7	14.06	95.8
5	558.7	95.5	13.72	93.5